

Dynamic fluid flow model for phosphate slurry pipeline: OCP main pipeline as case study

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ABSTRACT

The transportation of phosphate slurry through large-scale pipelines presents significant challenges due to the complex behavior of multiphase flows, particularly with varying solid content, density, and dynamic viscosity. Efficient and accurate prediction of flow behavior is critical for optimizing the operation of such pipelines. This work aims to develop a dynamic computational model to simulate phosphate slurry flow in pipelines. The case study focuses on the OCP Group's main slurry pipeline, which links the mining sites at Khouribga to the industrial plants at Jorf Lasfar, Morocco. This pipeline system spans a total length of 187.124 km, consisting of 5237 pipes with an inner diameter ranging from 0.8546 m to 0.8578 m, and features several elevation changes along its ground level. Using a section-averaged, dynamic approach and the Finite Volume scheme, the model computes essential flow parameters, including density, dynamic viscosity, and pressure, for the incompressible and non-Newtonian slurry and process water flows. The model's accuracy is validated against on-site measured data, showing an average deviation of $\pm 0.32\%$ for outlet density, and below 10% for pressure along the pipeline, underscoring the model's reliability. Additionally, a sensitivity analysis was conducted to illustrate the impact of key parameters on the predicted head losses and pressures along the pipeline. This analysis shows that the slurry viscosity is the most critical parameter, significantly influencing these predictions. This model provides high accuracy and reasonable CPU time for real-time simulation and monitoring, while also offering significant potential for optimizing pipeline operations and ensuring the reliability of phosphate transport.

1. Introduction

1.1. Industrial context

OCP Group is a worldwide industrial company, operating in the mining and marketing of the phosphate ore and manufacturing of derivatives, mainly fertilizers. The industrial process involves transporting the enriched ore from the mining sites to the processing plants. Since 2014, as part of a cost optimization strategy, OCP Group was decided to replace rail transportation with a continuous flow of phosphate slurry through a network of pipelines (see Fig. 1). This decision was primarily driven by economic factors, as rail transport is significantly more expensive and time-consuming compared to pipeline transport. The pipeline system not only reduces costs but also allows for a more efficient and continuous flow of phosphate slurry over long distances. Additionally, the use of pipelines minimizes the handling

and transportation losses that are often associated with rail transport. As a result, this transportation mode has proven to be very safe, economical, and environmentally friendly. The system consists now of four pipelines: the main pipeline extends over 187.127 km of land with highly variable topography (Rusconi et al., 2016); the Beni-Amir pipeline over 14.909 km with a transport capacity of 1500 t/h; the Sidi-Daoui pipeline over 22.262 km with a transport capacity of 1400 t/h; and the Merah El Ahrach (MEA) pipeline over 1.965 km and with transport capacity of 1400 t/h for low grade quality (MEA 12) and 700 t/h for high-grade phosphate quality (MEA 16). Recent studies have been devoted to this network of pipelines. El Hamra et al. (2022) have investigated a part of the main pipeline (over 6 km) to calculate the pressure loss, using the Finite Volume Method with a non-Newtonian flow model. Elkari et al. (2023a,b) have developed a surrogate model for pressure loss predictions, using a full three-dimensional, CFD model

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Nomenclature

(x, y, z)	Cartesian coordinates
ΔP	Pressure drop, [Pa]
A	Pipeline section area, [m ²]
D	Diameter, [m]
d	Particle size diameter, [μm]
f	sigmoid function
f_D	Darcy friction factor
g	Gravity acceleration, [m/s ²]
I	Momentum, [kg m/s]
L	Pipe length, [m]
m	Mass, [kg]
N	Number of equidistant points at the finite volumes
P	Hydro-static pressure, [Pa]
q	Flow rate, [m ³ /h]
Re	Reynolds number
S	Section, [m ²]
t	Time, [s]
U	Average velocity over the section, [m/s]
u	Average velocity, [m/s]
$\mathbf{n}$	Normal vector pointing
$\mathbf{u}$	Velocity vector

Greek symbols

α	volume fraction
λ	sigmoid parameter
μ	Dynamic viscosity of the mixture, [Pa s]
Π	Pressure, [Pa]
ρ	density of the mixture, [kg/m ³]
θ	Inclination angle, [deg]
ε	roughness, [mm]
φ	Curvature loss coefficient
φ_i	Mass flux, [kg/s]

Subscripts

in	Inlet flow
out	Outlet flow
p	Phosphate phase
s	Section
sv	Settling velocity
w	Process-Water phase

Acronyms

$F.V.M.$	Finite Volume method
$H.S.$	Head station
$I.Q.R.$	Inter-Quartile Range method
$P.M.S.$	Pressure Monitoring Sensors
$V.S.$	Valve station

and considering Eulerien two-phase flows models within a unit-length lab scale horizontal pipeline.

1.2. Short review of slurry modeling

Flow models are necessary to predict such system behavior to ensure optimal control and to prevent operating issues such as ore deposition, slug flow, pressure drops, and damages that can result from such events. In this context, two main categories of study can be

distinguished: the first concerns turbulent slurry flow modeling, and the second focuses on the rheological behavior and phenomenology of the slurry. The two are linked as the turbulent flow studies depend heavily on modeling the slurry's rheology. In [Jacobs \(1991\)](#), the authors studied the transport of solid-liquid phases, which is also called slurry transport in [Durand \(1953\)](#) and [Wasp et al. \(1970\)](#). These authors developed a methodology to compute and predict hydraulic gradients for coarse particle slurry flows. A single set of conservation equations was used to describe the slurry flow, and empirical correlations were proposed for the pressure gradient. [Doron et al. \(1987\)](#) developed a two-layer model that takes into account the possibility of a moving or stationary lower layer (bed) for the prediction of flow patterns and pressure losses in the pipelines. Significant challenges occur when the particle phase is considered and modeled as a continuous phase. This special approach was performed in the past for gas-solid phases. In this context, [Gidaspow \(1994\)](#) considered empirical expressions for the solid-liquid phases, and [Lun et al. \(1984\)](#) and [Louge et al. \(1991\)](#), and more recently ([Gómez González and Garzó, 2022](#)) developed the kinetic theory of granular flow, which allows for comparison with single-phase fluid mechanics and the introduction of new dependent parameter, namely the volume fraction of the dispersed phase. [Roco and Cader \(1990\)](#) considered multi-sized solid particles and different densities to develop a computational model for the wear rate distribution in slurry pipeline cross-sections. One of the first experimental measurements of pressure drop and heat transfer was conducted by [Plass and Molerus \(1974\)](#) and [Zisselmar and Molerus \(1979\)](#) for solid-liquid turbulent flows. The authors measured the effect of the solid fraction, i.e., solid concentration, on turbulence intensity, heat transfer, and pressure drop through a horizontal narrow pipe. [Matousek \(2002\)](#) and [Vlasák et al. \(2019\)](#) conducted laboratory experiments using sand-water mixture flows through horizontal, vertical, and inclined pipes. They observed flow pattern variation and analyzed the flow characteristics for fully stratified and suspended flows. [Carravetta et al. \(2016\)](#) conducted experimental measurements of the pressure loss in slurry pipe flows consisting of a mixture of water and a uniform sediment. Four models were considered for the prediction of pressure losses due to friction, and the authors provided evidence for the accuracy of each model. From these works, it is clear that experimental measurements of pressure, density, and solid particle volume fraction are very difficult to obtain at all points and cross-sections along the pipe length, but it is easier to obtain them through numerical investigation. One of the important numerical studies was conducted by [Ling et al. \(2003\)](#). They used an Algebraic Slip Mixture model with the RNG $k - \varepsilon$ turbulent model to investigate the three-dimensional flow of silica sand-water slurry, and found that the mean slurry velocity was more important than settling velocity of solid particulates. Most pipeline numerical computations have focused on solid-liquid flow elbows with different geometries, as bends are the most problematic components in a hydraulic conveying pipeline system. [Zhou et al. \(2021\)](#) used Computational Fluid Dynamics (CFD) and Discrete Element Method to study the flow of coarse slurry particles through a bend with 5% solid friction at three different elbow orientations: 0°, 45° and 90°. From this study, different flow regimes were involved depending on pipe orientation and the slurry conveying velocity, and it was noted that the pressure drop and the erosion rate are lowest for the vertical-horizontal bend (i.e., 90°). Additionally, pressure losses were negligible regardless of the pipe configuration at high conveying slurry velocities. [Dong et al. \(2022\)](#) and [Singh et al. \(2021\)](#) provided more detailed information on the effect of velocity, solid friction, and geometry rates on the pressure drop. These authors focused on the influence of particle size in different slurries through a 90-degree elbow, finding that the head loss increased with the slurry velocity and solid concentration, and that the wall wear rate increased greatly with increasing velocity, solid concentration, and particle size. The outer wall of the bend experienced greater wear than other parts, and with increasing velocity, the settlement wear at the bottom of the pipe shifted to the outer wall of the pipe. [Niezgoda-Żelasko and](#)

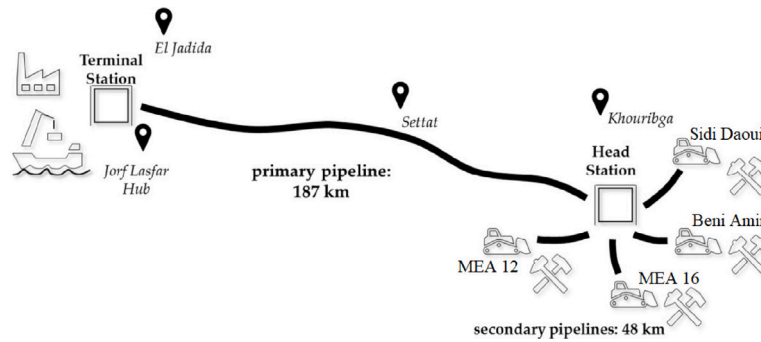


Fig. 1. The main and secondary pipelines between Khouribga and Jorf Lasfar (Geissler et al., 2018).

Zalewski (2006) and Wang et al. (2013) studied ice slurry flow in various pipes and compared a single-phase model (mixture model) and an Eulerian–Eulerian model. They found that the Eulerian–Eulerian model was the most accurate for computing the velocity and the pressure drop, using experimental comparisons. Lagrangian models have also been used to simulation particle-fluids pipe systems with varying slopes (Benjelloun et al., 2012) but are limited by their computing cost. Grozdek et al. (2009) investigated the relationship between the pressure drop and the particle volume fraction and presented a comparison between experimental and analytical results based on Poiseuille and Buckingham-Reiner models for ice slurry flow in a horizontal pipe. Their results focused on predicting the friction coefficient and hydraulic gradient as a function of ice solid concentration. They found also that the pressure drop in the turbulent regime was higher than in a single-phase flow. Liu et al. (2021) used Granular Dynamic Theory to give an accurate correlation between the dynamic viscosity of ice slurry and average velocity of the flow, while Bordet et al. (2018) conducted experiments in a horizontal pipe and found that ice slurry pressure drop was much more significant at high ice volume fractions and decreased velocity, and was higher than that for pure fluid. Shi et al. (2022) conducted a high-pressure flow loop experiment to study the migration and formation of micron-sized sand particles in slurry flow. This work can be complemented by the study of Sun et al. (2022), which focuses on characterizing the rheology of methane hydrate slurry in a continuous water within a horizontal system. Further research about multiphase slurries flows composed of different solid particles through horizontal, vertical or inclined pipes are reviewed in the work of Messa et al. (2021) and Elkarii (2023). The authors discussed the complexity of modeling slurry transport in large-scale pipelines stems from the intricate interplay of several physical phenomena. These include the turbulent nature of the flow, buoyancy effects due to density differences, and the interactions between particles as well as between particles and the pipe walls. Such interactions involve lubrication and collision forces, which significantly impact the overall flow behavior. Furthermore, simulating these dynamics accurately while ensuring computational efficiency presents an additional challenge due to the substantial computational time required for large-scale pipeline systems.

Another important aspect to consider in the study of slurry transport is the rheology of the phosphate slurry. Previous studies, such as those conducted by Belbsir et al. (2018) and El Moçayd and Seaid (2021), have investigated the rheology of Moroccan phosphate slurries and proposed preliminary models for estimating the dynamic viscosity. However, a complete and accurate model or correlation to determine the dynamic viscosity of Moroccan phosphate slurries has not yet been developed.

1.3. Present study

To overcome the limitations of the previously cited experimental studies, which were restricted to small-scale and small-diameter

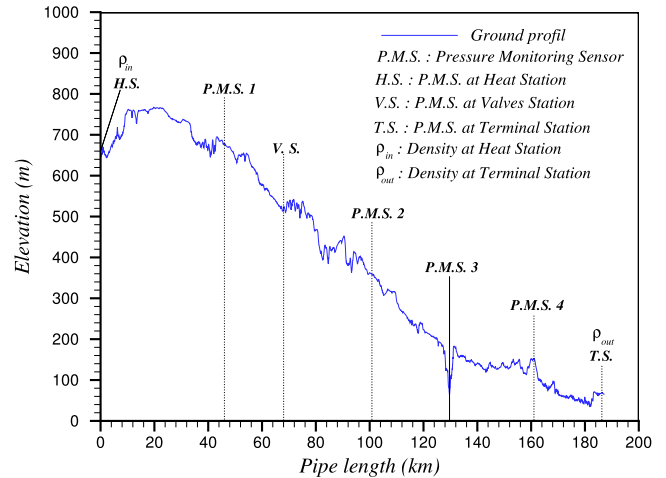


Fig. 2. Ground level of the slurry main pipeline.

pipes, or numerical models constrained to short-duration simulations, this study considers the actual pipeline geometry. Moreover, rather than focusing exclusively on continuous slurry flow, this research accounts for real-world operational scenarios, including both continuous batch slurry flows, as well as programmed and abrupt downtimes. The data covers approximately several days of operation, while ensuring minimal computation time (around twenty minutes).

Our current study focuses on developing a dynamic numerical model that enables to predict the flow characteristics of phosphate slurry at a large-scale. The model utilizes the actual and current geometrical and operational conditions for the phosphate slurry main pipeline, which is consisting of 5237 pipes and is constructed with the DN900 steel HDPE lined material. The inner diameter of the pipes is ranging from 0.8546 m to 0.8578 m. It should be noted that despite the small differences in the inner diameters of the pipeline, taking them into account is necessary as they have a relatively significant effect on the velocity. Fig. 2 represents the ground level variations along the total length of the slurry main pipeline (187.124 km). It is clear that the pipes slope varies significantly segmentation will be reproduced exactly in our numerical scheme. Fig. 2 also shows the positions of the seven pressure monitoring sensors and the two density monitoring sensors, whose measures will be used to validate our model.

1.4. Content

Section 2 presents the derivation of a section averaged model, allowing to represent the large scale industrial pipeline with tractable computational models. This derivation is applied for each reach with a constant inclination. Section 3 presents the finite volume approach

employed to solve the obtained Navier–Stokes-type equation, accounting for mass balance and particle transport equation. In this section, the initial and limit conditions are reproduced from the real and practice conditions. To implement the obtained model in the industrial full scale context, Section 4 is devoted to the general processing of the model input data (i.e. operating conditions) from noisy industrial sensors data. In Section 5, we present a use case for the main slurry pipeline. Section 6 is dedicated to the obtained numerical results and the comparison with historical measurements to validate the accuracy of the proposed model. In the same section, an sensitivity analysis of the numerical results is carried out. Finally, the main conclusions of the present study are drawn in Section 7.

2. Mathematical model

2.1. Governing equations

The governing equations, for the multi-dimensional model describing the isothermal flow of the non-Newtonian and incompressible slurry consist on mass and momentum conservation equations. Based on an Eulerian mixture approach, the two equations, in conservation form, read:

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) = 0, \quad (1)$$

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \operatorname{div}(\rho \mathbf{u} \otimes \mathbf{u}) + \nabla P = \rho \mathbf{g} + \operatorname{div}(\mu \nabla \mathbf{u}), \quad (2)$$

where ρ is the density of the mixture of phosphate slurry and process-water, $\mathbf{u}$ the velocity vector of the fluid, P the hydro-static pressure, μ the dynamic viscosity of the mixture, and $\mathbf{g}$ the gravity acceleration vector. The flow incompressibility hypothesis be written as:

$$\operatorname{div}(\mathbf{u}) = 0. \quad (3)$$

The density of the phosphate slurry is expressed as functions of the volume fraction α of solid phosphate particles, as follows:

$$\rho = \alpha \rho_p + (1 - \alpha) \rho_w \quad (4)$$

where subscripts p and w represent the solid phosphate and process-water phases.

In the context of the industrial pipeline operations, the boundary conditions are specified only as section's averages. For the inlet section S_{in} , the average density and mass flow are given by:

$$\begin{cases} \rho_{in}(t) = \frac{1}{\mathcal{A}} \int_{S_{in}} \rho(0, \sigma, t) d\sigma, & (a) \\ q_{in}(t) = \int_{S_{in}} \rho(0, \sigma, t) \mathbf{u}(0, \sigma, t) \cdot \mathbf{e}_x d\sigma & (b), \end{cases} \quad (5)$$

with $\mathcal{A}$ the pipeline section area. At the outlet, the average pressure P_{out} is monitored and defined by:

$$\frac{1}{\mathcal{A}} \int_{S_{out}} P(L_x, \sigma, t) d\sigma = P_{out}(t) \quad (6)$$

The roughness of the inlet wall boundary is considered constant and assumed to be $\varepsilon = 0.0213$ mm in the application presented in this paper. A no-slip condition is hence considered on the internal wall of the pipeline $u = 0$. We will also consider the flow insurance conditions, i.e. that the section S_{in} is actually an inlet

$$\int_{S_{in}} \rho \mathbf{u} \cdot \mathbf{n} d\sigma < 0, \quad \forall t \geq 0 \quad (7)$$

with $\mathbf{n}$ the normal pointing towards the pipe exterior. Similarly, at the outlet section :

$$\int_{S_{out}} \rho \mathbf{u} \cdot \mathbf{n} d\sigma > 0 \quad \forall t \geq 0 \quad (8)$$

Given the condition (7) with $\mathbf{n} = -\mathbf{e}_x$ on S_{in} we have

$$q_{in}(t) > 0, \quad \forall t \geq 0 \quad (9)$$

We can define the average inlet velocity as follows:

$$v_{in}(t) = \frac{\int_{S_{in}} \rho \mathbf{u} \cdot \mathbf{e}_x d\sigma}{\int_{S_{in}} \rho d\sigma} = \frac{q_{in}(t)}{A \rho_{in}(t)} > 0 \quad (10)$$

2.2. Consequences of the incompressibility hypothesis

Let us consider a reach T of the pipe inclined at an angle θ relative to the horizontal with a local frame (O, x, y, z) as depicted in Fig. 3. Denoting D as the inner pipe diameter and L its length.

$$T = \{(x, y, z), \quad y^2 + z^2 < D^2, \quad 0 < x < L\} \quad (11)$$

$$C = \{(x, y, z), \quad y^2 + z^2 = D^2, \quad 0 \leq x \leq L\} \quad (12)$$

$$S_{in}^T = \{(0, y, z), \quad y^2 + z^2 \leq D^2\} \quad (13)$$

$$S_{out}^T = \{(L, y, z), \quad y^2 + z^2 \leq D^2\} \quad (14)$$

The velocity $\mathbf{u} : \bar{T} \times \mathbb{R}_+ \rightarrow \mathbb{R}^3$, satisfies the incompressibility hypothesis (3), and we note v its x component along the pipe axis :

$$v(x, y, z, t) \equiv \mathbf{u}(\mathbf{x}, \mathbf{y}, \mathbf{z}, t) \cdot \mathbf{e}_x \quad (15)$$

We introduce the average velocity U for v over the sections $S_x = \{(x, y, z), \quad y^2 + z^2 \leq D^2\}$:

$$U(t, x) = \frac{1}{\mathcal{A}} \int_{S_x} v(x, y, z, t) d\sigma \quad (16)$$

We have the proposition :

Proposition 1. Under the above hypothesis (3) :

$$U(t, x) = \frac{1}{\mathcal{A}} \int_{S_x} v(x, y, z, t) d\sigma \quad \text{is independant of } x, \text{ i.e. } U(t, x) = U(t) \quad (17)$$

Proof. Let us integrate $\operatorname{div}(\mathbf{u})$ over the volume $V_\xi = \{(x, y, z), \quad 0 < x < \xi, \quad y^2 + z^2 < D^2\}$.

$$\int_{V_\xi} \operatorname{div}(\mathbf{u}) dx dy dz = \int_{\partial V_\xi} \mathbf{u} \cdot \mathbf{n} d\sigma, \quad \forall \xi > 0 \quad (18)$$

Given the boundary condition $u = 0$ on C , we get :

$$\int_{V_\xi} \operatorname{div}(\mathbf{u}) d\tau = \int_{S_{in} \cup S_\xi} \mathbf{u} \cdot \mathbf{n} d\sigma = - \int_{S_{in}} v d\sigma + \int_{S_\xi} v d\sigma \quad (19)$$

Hence, since $\operatorname{div}(\mathbf{u}) = 0$ in V_ξ , we have:

$$U(t, \xi) = \frac{1}{\mathcal{A}} \int_{S_\xi} v(x, y, z, t) d\sigma = \frac{1}{\mathcal{A}} \int_{S_{in}} v(0, y, z, t) d\sigma = U(t, 0) = U(t) \quad (20)$$

independently of ξ , which concludes the proof of Proposition 1 .

It should be noted that the inner diameter and section of the pipe changes slightly at some reaches T' . The axial velocity $U'(t)$ in T' can then be calculated based on flow conservation between the two sections $\mathcal{A}$ and $\mathcal{A}'$ as follows: $U'(t) = U(t) \left(\frac{\mathcal{A}}{\mathcal{A}'} \right)$. The above considerations will allow us the define a tractable section averaged finite volume computational model, that is in addition in coherence with the operations boundary conditions (5a)–(5b)–(6).

Finally, it is important to note that in this model, the settling of solid particles is considered negligible. In fact, Ling et al. (2003) introduced a simplified three-dimensional algebraic slip mixture (ASM) model for the numerical simulation of sand–water slurry flows. This ASM model

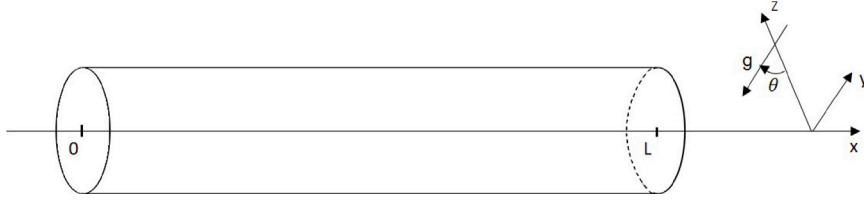
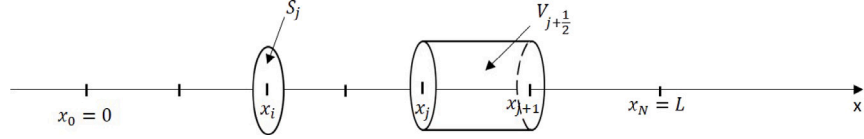
Fig. 3. A reach/segment T of the pipe with constant inclination θ .

Fig. 4. 1D finite volume discretization.

was coupled with the renormalization group *RNG* $K-\epsilon$ turbulence model to solve fully developed turbulent flows. Their findings suggest that the slip velocity magnitude is typically small, making its effect on slurry flow negligible in both heterogeneous and sliding bed flow for volumetric concentrations up to an average of 20% (in this study, it ranged from 0.7 to 30%). Moreover, this assumption is strongly validated when the average slurry velocity (2.2, m/s, as observed in the present study) is greater than the settling velocity (6.86×10^{-2} , m/s for phosphate particles with a diameter, <500 , μm , see [Annexe](#)). On the other hand, it should be noted that the present model, as will be discussed in Section 3, is limited to a 1D cross-sectional averaged approach, developed by averaging the flow properties over the pipe's cross-section and integrating them along its longitudinal axis.

3. Section averaged finite volume scheme

We consider again a segment T (see [Fig. 3](#)) of length L of the pipeline, and a uniform 1D meshing with $N \geq 1$ finite volumes $V_{i+1/2}$, for $0 \leq i \leq N$. We set $x_i = \frac{iL}{N}$, $0 \leq i \leq N$ the $N+1$ equidistant points at the finite volumes boundaries, as illustrated in [Fig. 4](#). We denote by:

$$\begin{cases} S_i = S_{x_i} = \{(x_i, y, z), \quad y^2 + z^2 \leq D^2\} & \text{(a)} \\ V_{i+1/2} = \{(x, y, z), \quad x_i < x < x_{i+1}, \quad y^2 + z^2 < D^2\} & \text{(b)} \end{cases} \quad (21)$$

Integrating (1) over $V_{i+1/2}$ gives :

$$\frac{d}{dt} \int_{V_{i+1/2}} \rho d\tau + \int_{S_{i+1}} \rho v dC - \int_{S_i} \rho v d\sigma = 0, \quad 0 \leq i \leq N-1 \quad (22)$$

Integrating (2) over $V_{i+1/2}$, and projecting over the x axis gives :

$$\begin{aligned} \frac{d}{dt} \int_{V_{i+1/2}} \rho v d\tau + \int_{S_{i+1}} \rho v^2 d\sigma - \int_{S_i} \rho v^2 d\sigma + \int_{S_{i+1}} P d\sigma - \int_{S_i} P d\sigma = \\ \int_{V_{i+1/2}} \rho \mathbf{g} \cdot \mathbf{e}_x d\tau + \int_{V_{i+1/2}} \mu \Delta v d\tau \end{aligned} \quad (23)$$

The above system (22)–(23) is an integro-differential system for the $2N$ finite volumes variables:

$$m_{i+1/2} \equiv \int_{V_{i+1/2}} \rho d\tau \quad 0 \leq i \leq N-1 \quad (24)$$

$$I_{i+1/2} \equiv \int_{V_{i+1/2}} \rho v d\tau \quad 0 \leq i \leq N-1 \quad (25)$$

where the mass and the momentum $\left(m_{i+1/2}, I_{i+1/2}\right)_{i=0}^{N-1}$ depending only on time.

3.1. Mass conservation

The mass conservation Eq. (22) can be written as:

$$\frac{dm_{i+1/2}}{dt} = \int_{S_i} \rho v d\sigma - \int_{S_{i+1}} \rho v d\sigma, \quad 0 \leq i \leq N-1 \quad (26)$$

Thus, we need to approximate the $N+1$ fluxes:

$$\varphi_i \equiv \int_{S_i} \rho v d\sigma, \quad 0 \leq i \leq N \quad (27)$$

We note that for $i = 0$: $\varphi_0(t) = \int_{S_0} \rho v d\sigma = q_{in}(t)$ by Eq. (5b). To determine $\varphi_1, \dots, \varphi_N$, the fundamental remark is that, using [Proposition 1](#), the average velocity on the section S_i may be written as:

$$U_i(t) = \frac{1}{A} \int_{S_i} v(x, y, z, t) d\sigma = U_0(t) = \frac{1}{A} \int_{S_{in}} v(0, y, z, t) d\sigma \quad (28)$$

Given that $\int_{S_0} \rho v d\sigma = q_{in}(t) > 0$, we assume $U_0(t) > 0, \quad \forall t \geq 0$. Hence, the upwind approximation for φ_i is :

$$\varphi_i = \int_{S_i} \rho v d\sigma \simeq \frac{A}{V_{i-1/2}} \int_{V_{i-1/2}} \rho v d\tau = \frac{1}{A \Delta x} I_{i-1/2} = \frac{I_{i-1/2}}{\Delta x} \quad (29)$$

with $\Delta x = \frac{L}{N}$. Hence Eq. (26) is approximated as :

$$\frac{dm_{i+1/2}}{dt} = \frac{I_{i-1/2} - I_{i+1/2}}{\Delta x}, \quad 0 \leq i \leq N-1 \quad (30)$$

with $I_{-1/2} \equiv \Delta x q_{in}(t)$ given by the boundary conditions. On the other hand, we note that :

$$\frac{I_{i+1/2}}{m_{i+1/2}} = \frac{\int_{V_{i+1/2}} \rho v d\tau}{\int_{V_{i+1/2}} \rho d\tau} = \langle v \rangle_{\rho, V_{i+1/2}}, \quad \text{for } 0 \leq i \leq N-1 \quad (31)$$

According to [Proposition 1](#), we know that $U_i = \langle v \rangle_{S_i}$ remains constant with respect to i . Consequently, we will make the following approximation:

$$\langle v \rangle_{\rho, V_{i+1/2}} \simeq \langle v \rangle_{S_i} = U(t) = v_{in}(t) \quad (32)$$

As a result, Eq. (31) transforms to:

$$\frac{I_{i+1/2}}{m_{i+1/2}} = v_{in}(t) \equiv \frac{q_{in}}{A \rho_{in}} \quad (33)$$

Subsequently, Eq. (30) is further simplified to:

$$\frac{dm_{i+1/2}}{dt} = \frac{v_{in}}{\Delta x} (m_{i-1/2} - m_{i+1/2}), \quad 0 \leq i \leq N-1 \quad (34)$$

with the boundary condition: $m_{-\frac{1}{2}} \equiv \frac{\Delta x q_{in}}{v_{in}} = \mathcal{A} \Delta x \rho_{in}$. Once $m_{i+\frac{1}{2}}$ are computed, $I_{i+\frac{1}{2}}$ can then be computed as :

$$I_{i+\frac{1}{2}} = v_{in}(t) m_{i+\frac{1}{2}} \quad (35)$$

Hence in the above, we used the mass conservation as well as the incompressibility hypothesis to compute $m_{i+\frac{1}{2}}$ and $I_{i+\frac{1}{2}}$. The momentum equation can then be used to compute the pressures, which we will do in the following subsection.

3.2. Discrete momentum conservation

To approach the convection term in the momentum Eq. (23), we use again Proposition 1 to write for $0 \leq i \leq N$:

$$\int_{S_i} \rho v^2 d\sigma \simeq U_i \int_{S_i} \rho v d\sigma = v_{in}(t) \int_{S_i} \rho v d\sigma = v_{in}(t) \frac{I_{i-\frac{1}{2}}}{\Delta x} \quad (36)$$

Furthermore, we note

$$\Pi_i \equiv \frac{1}{\mathcal{A}} \int_{S_i} P d\sigma, \quad 0 \leq i \leq N \quad (37)$$

From (6), we have $\Pi_N = P_{out}(t)$. The gravitational contribution is represented by

$$\int_{V_{i+\frac{1}{2}}} \rho \mathbf{g} \cdot \mathbf{e}_x d\tau = -|g| \sin(\theta) \int_{V_{i+\frac{1}{2}}} \rho d\tau = -|g| \sin(\theta) m_{i+\frac{1}{2}}$$

The momentum equation Eq. (23) writes then as:

$$\frac{dI_{i+\frac{1}{2}}}{dt} + \frac{v_{in}}{\Delta x} (I_{i-\frac{1}{2}} - I_{i+\frac{1}{2}}) + \mathcal{A}(\Pi_i - \Pi_{i+1}) = -|g| \sin(\theta) m_{i+\frac{1}{2}} + \int_{V_{i+\frac{1}{2}}} \mu \Delta v d\tau$$

Hence, we need to model the terms $\int_{V_{i+\frac{1}{2}}} \mu \Delta v d\tau$. We use a pressure loss model of the form

$$\int_{V_{i+\frac{1}{2}}} \mu \Delta v d\tau = \mathcal{A} \Delta P_{i+\frac{1}{2}} = \mathcal{A}(\Delta P)_{Fr,i+\frac{1}{2}} \quad (38)$$

with $\Delta P_{i+\frac{1}{2}} = (\Delta P)_{Fr,i+\frac{1}{2}}$ is the pressure loss throughout the cell, due to friction. $(\Delta P)_{Fr,i+\frac{1}{2}}$ depends on the length Δx and the fluid properties at the finite volume $V_{i+\frac{1}{2}}$ such as the average viscosity $\mu_{i+\frac{1}{2}}$, and different pressure-loss laws can be used as we will discuss in Section 3.3. Now, Eq. (23) is approximated by :

$$\frac{dI_{i+\frac{1}{2}}}{dt} = \frac{v_{in}}{\Delta x} (I_{i-\frac{1}{2}} - I_{i+\frac{1}{2}}) + \mathcal{A}(\Pi_i - \Pi_{i+1}) - |g| \sin \theta m_{i+\frac{1}{2}} + \mathcal{A} \Delta P_{i+\frac{1}{2}} \quad (39)$$

Hence, noting that from Eq. (35) we have

$$\frac{dI_{i+\frac{1}{2}}}{dt} = v_{in} \frac{dm_{i+\frac{1}{2}}}{dt} + m_{i+\frac{1}{2}} \frac{dv_{in}}{dt} \quad (40)$$

we get :

$$\Pi_i = \Pi_{i+1} + \frac{1}{\mathcal{A}} \left(|g| \sin \theta + \frac{dv}{dt} \right) m_{i+\frac{1}{2}} + \Delta P_{i+\frac{1}{2}}, \quad 0 \leq i \leq N-1 \quad (41)$$

with the boundary condition $\Pi_N = P_{out}(t)$, given by Eq. (6). Rearranging this system of equations, we obtain :

$$\Pi_j = \Pi_N + \frac{1}{\mathcal{A}} \left(\sum_{i=j}^{N-1} m_{i+\frac{1}{2}} \right) \left(\frac{dv_{in}}{dt} - |g| \sin(\theta) \right) - \sum_{i=j}^{N-1} \Delta P_{i+\frac{1}{2}} \quad (42)$$

3.3. Pressure loss models and contact zones

We present here an example model for the pressure losses due to the friction. In our case, several pressure loss models have been implemented, including ones base on full 3D numerical models (Elkari et al., 2023a). For instance, the pressure drop ΔP can be determined based on the Darcy-Weisbach equation for an incompressible fluid,

including the linear and singular (i.e. in bending parts of the pipe) contributions:

$$\begin{cases} \Delta P_{i+\frac{1}{2}} &= \Delta P_l + \Delta P_s \\ \Delta P_{linear} &= \Delta x \rho_{i+\frac{1}{2}} \frac{f_D}{2D} v_{i+\frac{1}{2}}^2 \\ \Delta P_{singular} &= 0.0033 \varphi \frac{1}{2} \rho_{i+\frac{1}{2}} v_{i+\frac{1}{2}}^2 \end{cases} \quad (43)$$

where φ is the curvature loss coefficient ($= 0$ for a straight pipe) and f_D is the Darcy friction factor. For the specific case of the phosphate slurry, and as discussed in Idelchik (1986), f_D is given by:

$$f_D = \frac{a}{\left[\log \left(\frac{\epsilon}{3.7D} + \frac{5.74}{Re^{0.9}} \right) \right]^2} \quad (44)$$

with $Re = \frac{\rho v D}{\mu_{i+\frac{1}{2}}}$ is the local Reynolds number. To compute the viscosity at different locations of the pipe, a transport equation for the dynamic viscosity, along with the known viscosity at the inlet boundary are used ($U(t)$ is defined in (32))

$$\begin{cases} \frac{\partial \mu}{\partial t} + U(t) \frac{\partial \mu}{\partial x} &= 0 \\ \mu(x=0, t) &= \mu_{in}(t) \end{cases} \quad (45)$$

with $\mu_{in}(t)$ is the measured dynamic viscosity at the inlet of the main pipeline.

The density and dynamic viscosity in the contact zone between phosphate slurry and process water batches change drastically. A relatively thin diffusion interface is present around the contact zone. To model this interface, a sigmoid function (Han and Moraga, 1995) is considered in order to interpolate the density and the dynamic viscosity values. For two consecutive times (t_a, t_b) , where the density jumps from ρ_a to ρ_b and dynamic viscosity from μ_a to μ_b , where subscripts (a, b) represent phosphate slurry and process water respectively, the density and dynamic viscosity $f = (\rho \text{ or } \mu)$ can be determined as follows:

$$\begin{cases} f^{t_a+\Delta t} &= f_a + \frac{f_b - f_a}{1 + \exp\left(\frac{0.5(t_a + t_b) - t}{\lambda(t_b - t_a)}\right)} & \text{if } f_a \neq f_b \\ f^{t_a+\Delta t} &= f_a & \text{if } f_a = f_b \end{cases} \quad (46)$$

The sigmoid parameter $\lambda = 0.1$ is calibrated from process data, for time intervals such as $\Delta t = t_b - t_a \leq 2$ s.

4. Model input data and solution procedure

The dynamic and discrete governing equations (34)–(42)–(45) are implemented using the Modelica language and solved using the Open-Modelica solver. In addition to the geometrical parameters, the model's input data consists of boundary conditions specified in Eq. (5a)–(5b)–(6), which govern the temporal evolution of the inlet parameters, including density, volumetric flow rate, and dynamic viscosity. Furthermore, it encompasses the controlled output pressure at the Terminal Station.

Practically, we conduct a real-time simulation using the operational inlet data such as inlet density, inlet flow rate and outlet pressure, which are measured by field instruments on-site at the Head Stations (for inlet parameters) and Terminal Station (for outlet parameter), respectively. However, the series data from the sensors are very noisy, due to sensitivities of the sensors. Therefore, we used an Inter-Quartile Range (IQR) criterion to identify and remove outliers corresponding to aberrant data, and apply a Sigmoid function to smooth the time series over time.

Let us detail the IQR process. We initially segment the density sensors data into sub-groups over the total time range, where each sub-group represents a transported batch type : phosphate slurry or process water. After computing the second quartile value for each sub-group, the outliers values are replaced with this value.

The dynamic viscosity is measured for each batch of the phosphate slurry at the Head Station OCP Group laboratory. We recall that our

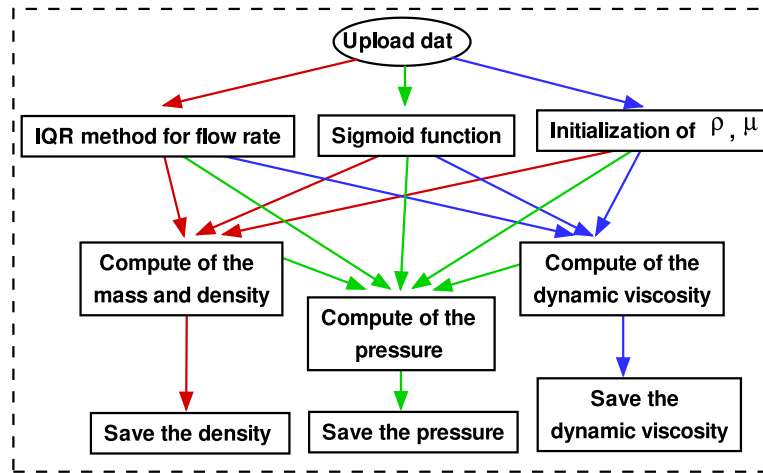


Fig. 5. The digital twin model flowchart.

computational model propagates this measured inlet viscosity within the pipe according to Eq. (45). Two parameters can affect the accurate measurement of the phosphate slurry dynamic viscosity. The first one is related to the method of sampling. Thus, the sampling is done in three periods for each batch, and an average dynamic viscosity is considered for the batch. The second one is related to the accuracy of the measurement technique at the OCP Group laboratory.

The input data processing, along with the numerical computation steps, are schematized in flowchart 5.

5. Slurry main pipe validation use case

In order to validate our numerical modeling, input data (i.e. the boundary conditions) in Eqs. (5a)–(5b)–(6) are taken from historical operational series data, and the simulation results from the model are compared to historical density and pressure measures from the several monitoring sensors (see Fig. 2).

5.1. Grid independence

For every single reach/segment of length L , and inclination angle θ , the governing equations (34)–(42)–(45) are discretized in space using $N = \frac{L}{\Delta x}$ finite volumes and in time with a time step Δt . The implicit numerical scheme which is adopted may be diffusive. To assess its accuracy and determine appropriate space and time mesh sizes for the numerical simulations, a grid independence test has been conducted with varying space and time step sizes. The density variable is chosen to carry out this test.

To evaluate grid independence, the relative error: $Error(\%) = \frac{\rho_{ref} - \rho}{\rho_{ref}} \times 100$ is computed relative to the results obtained with the finest space step size ($N = 1000$) and the smallest time step size ($\Delta t = 1$ s), which are considered as reference solutions ρ_{ref} .

The results are illustrated in Tables 1–2. Table 1 shows the effect of space step size on the outlet density for a fixed time step size of $\Delta t = 2$ s. The results indicate that as N increases, the error in the computed density decreases, suggesting convergence. The smallest error observed with $N = 500$ relative to the reference solution indicates that this mesh size provides a sufficiently fine discretization for accurate simulations. Similarly, Table 2 demonstrates the effect of time step size on the outlet density for a fixed space step size of $N = 500$. The deviation in density is compared to the reference solution with $\Delta t = 1$ s. The results show that $\Delta t = 2$ s provides a good balance between accuracy and computational efficiency, as larger time steps introduce significant deviations. Based on these analyses, the optimal space and time step sizes were determined to be $N = 500$ and $\Delta t = 2$ s, respectively. These values were chosen as they achieve a minimal relative error compared to the reference solutions while ensuring the accuracy of calculations.

Table 1

Effect of space step size on outlet density for $\Delta t = 2$ s.

N	ρ	Error (%)
10	1154.2484	4.3835
100	1112.2046	0.5813
500	1106.5219	0.0674
1000	1105.7766	–

Table 2

Effect of time step size on outlet density for $N = 500$.

Δt	ρ	Error (%)
1	1113.0181	–
2	1112.2046	0.0731
5	1166.1933	4.7776
10	1200.7003	7.8779

5.2. Input data analysis and pre-treatment

The results presented hereafter are based on recent and real measured data at OCP group, for an operating time of approximately 127 h. The collected data from the OCP Group Supervisory Control And Data Acquisition (SCADA) system represents 2553 data points, with a time step of 180 s. The solid particle sizes considered in this data ranges from 10 μm to 500 μm . The concentration of solid particles by volume is from 0.7 to 30%, and the measured velocity is from 0 to 3 m/s.

Fig. 6 displays the inlet flow rates obtained from the measuring instrument. It should be mentioned that the nominal accuracy of this flow meter is $\pm 1\%$. Fig. 6 shows that abrupt changes in the flow operating conditions occurs from high $Q \neq 0$ m³/h to very low $Q \approx 0$ m³/h flow rates, which is caused by unscheduled downtime (e.g. electricity cut-off), or scheduled downtime when transitions from the main pipe feeding train A to feeding train B occurs. These downtime periods are indicated in red line in Fig. 6 and subsequent Figures.

In order to single out outlier values, the Inter-Quartile Range method (IQR), presented in Section 4, is used. The obtained input flow rate is also illustrated in Fig. 6.

Fig. 7 illustrates the inlet density where two ranges of density are to be distinguished depending on the operating conditions. First, the pulp density is characterized by its high value $\rho > 1400$ kg/m³. The second one is related to the process-water flow $\rho = 1000$ kg/m³, which is also used when the pumping is stopped $Q \approx 0$ m³/h, in order to avoid clogging of solid particles inside the pumps, or as a transported batch to clean the pipeline and prevent particles clogging along it. The density is measured using the Laser hygrometer of an nominal accuracy of $\pm 1\%$.

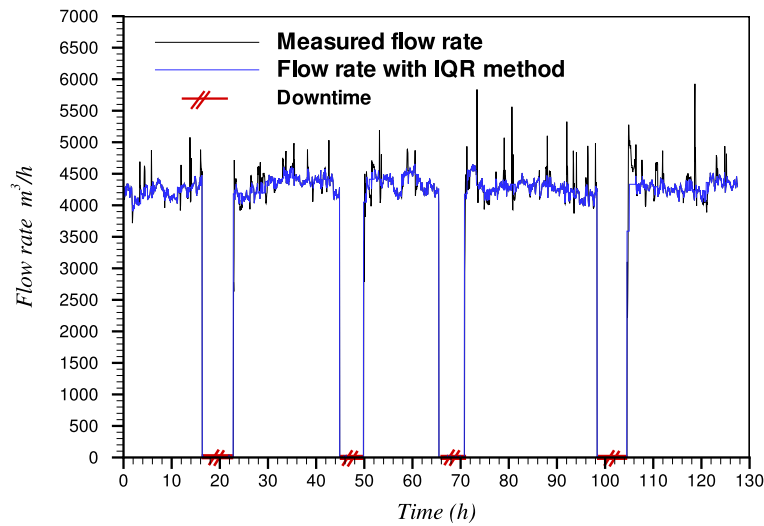


Fig. 6. Measured input flow rate from sensors and processed input flow rate with IQR method. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

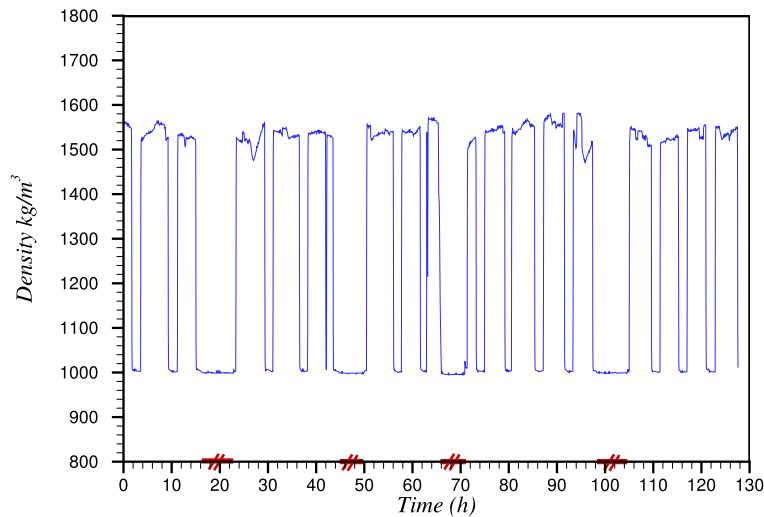


Fig. 7. Inlet density from operational data.

The inlet dynamic viscosity illustrated in Fig. 8 changes dependently on the chemical composition and the particle sizes distribution for each slurry batch.

6. Results and discussion

6.1. Simulation results

First, as model output, we consider the density at the outlet of the main pipeline (Terminal Station). The result can be compared to measured series data from the density meter at the outlet of the main pipeline. It should be noted that along the pipeline, only two density meters are available at its input and output. Fig. 9 illustrates a comparison between the measured outlet density values and those computed using the current numerical model. Two different ranges are to be distinguished, the first one, when the phosphate slurry or water process flows are continued, i.e., $Q > 0 \text{ m}^3/\text{h}$. In this situation, the obtained result shows that the outlet density calculated has the same behaviors than the measured density evolution with an average deviation of around $4.8 \text{ kg/m}^3 (\pm 0.32\%)$. This proves that the our numerical results are in good agreement with the experimental measurements for predicting the outlet density evolution. This also validates the

computed density at any position in the main pipeline. The second one is when the downtime flow occurs (time intervals in red line in Fig. 6) and as discussed previously, the sedimentation of the phosphate solid particles may increase (or decrease) the measured mixture density by the sensor, and as a result, an important difference is obtained with respect to the computed density values. In fact, we note that in the present simulations results, and according to Eqs. (34) and (45), when $V \rightarrow 0 \text{ ms}^{-1}$, the density and dynamic viscosity remain constants. The total number of data point excluding downtime periods is 2061 points (we recall the total data points number is 2553). Subsequently, we will evaluate our model results only on these 2061 continuous flow data points.

Fig. 10 illustrates a comparison between the measured pressure and the computed one. According to Fig. 2, in terms of pressure, there are six pressure monitoring sensors (PMS) available along the slurry main pipeline. For continuous fluid flow, where $Q \gg 0$, Fig. 10 shows that all computed pressures follow nearly the same trend as the measured ones. To provide a clearer visual comparison between measured and predicted pressure data, Fig. 11 is presented. From this figure, it is clear that there is an excellent agreement between experimental measurement data and predicted values, as there is a compact cluster of data points within the tolerance interval (red line), representing the

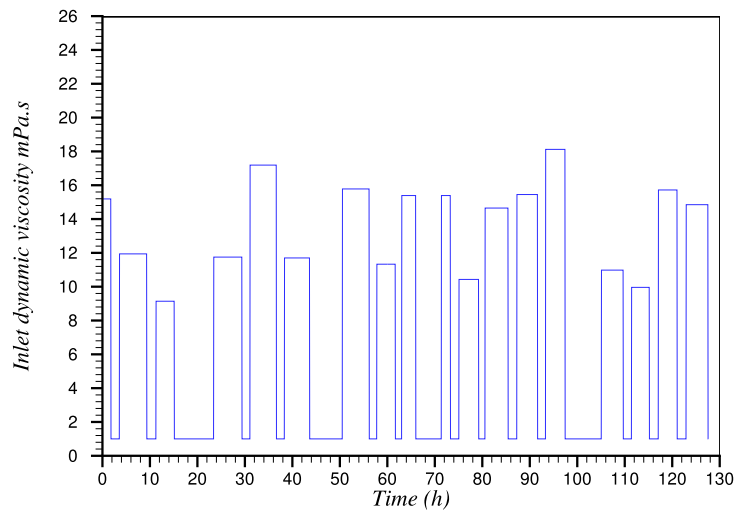


Fig. 8. Inlet dynamic viscosity from operational data.

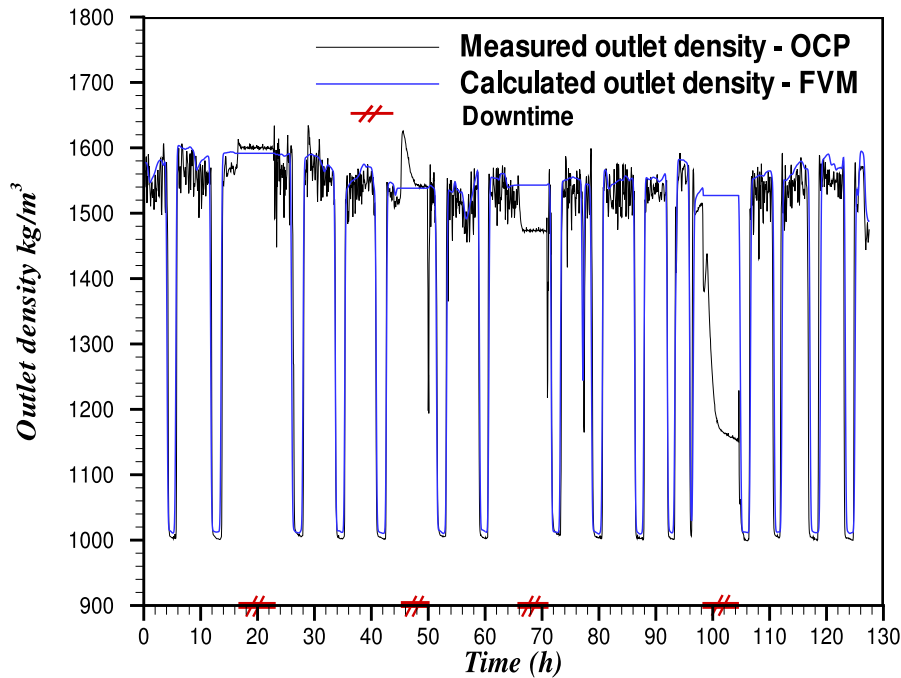


Fig. 9. Measured and computed outlet density.

root mean squared percentage difference (RMSPD). This indicates that the measured data and predicted values are appropriately close to each other. Although the relatively large deviations (15.21%) at the pressure monitoring sensor PMS-1 may be attributed to poor sensor calibration in the range of relatively low pressures observed at the PMS-1 location, RMSPD values for the other sensors, less than 10%, further confirm the accuracy and precision of the developed model in this study.

6.2. Sensitivity analysis

In this Section, we investigate the sensitivity of the model outputs (i.e., the inlet pressure and outlet density) to disturbances in input model parameters around nominal values. This analysis aims to evaluate the effect of measurement errors in these input model parameters. The model outputs are compared to measured data for a batch of

continuous phosphate slurry flow (interval [71 h–91 h] in the use case presented in the previous Section), with $N = 540$ data points.

Considering the effect of input flow rate variation ($\Delta Q \in [0.1\%, 10\%]$), Fig. 12 shows the Mean Absolute Percentage Error Eq. (48) and the corresponding pressure deviation over the indicated period. From this Figure, and assuming a 100 kPa pressure tolerance, it appears that the maximum permissible error in the volumetric flow rate value is $\pm 0.65\%$ which corresponds approximately to $108.51 \text{ m}^3/\text{h}$, in order to ensure the safety and integrity of the equipment and the accuracy of the predicted FVM model values.

$$MAPE = \frac{100}{N} \sum_{i=1}^N 2 \times \frac{|P_i^{\text{computed}} - P_i^{\text{measured}}|}{P_i^{\text{computed}} + P_i^{\text{measured}}} \quad (47)$$

Fig. 13 illustrates the mean absolute percentage error obtained when increasing the measured dynamic viscosity until 20%. The origin

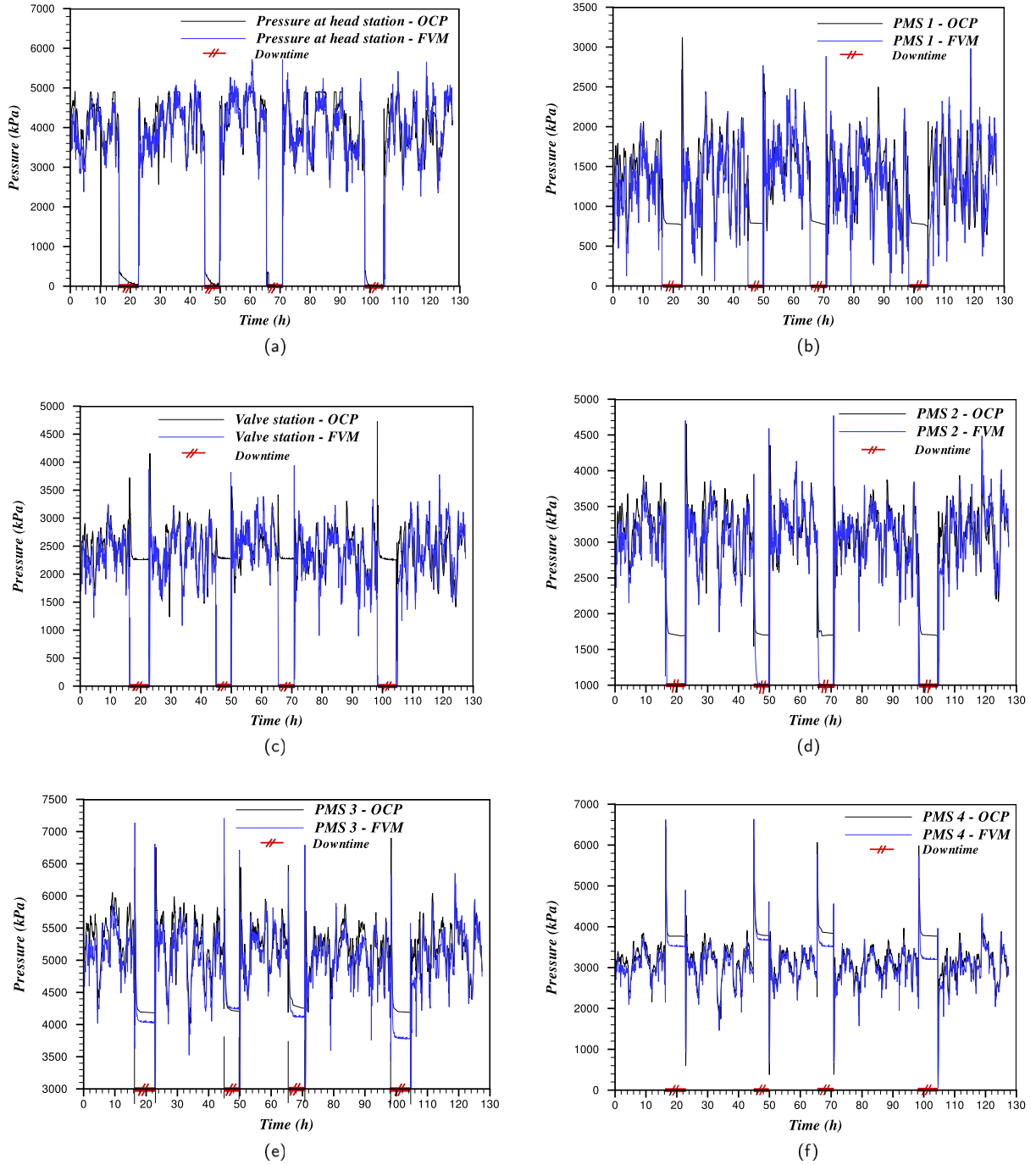


Fig. 10. Comparison between pressure monitoring sensor data and computed pressure at (a) Input main pipeline, (b) PMS-1, (c) Valve station, (d) PMS-2, (e) PMS-3 and (f) PMS-4.

of the inaccuracy in dynamic viscosity measured values is the experimental uncertainties of the measurement technique at the laboratory and the method of collecting the three samples from the same batch. From Fig. 13, it appears that an error of $\pm 7.29\%$ in the dynamic viscosity involves a 100 kPa deviation in the inlet pressure computed values.

The same analysis is carried out for the effect of the inlet density and outlet pressure. Figs. 14.a and .b reveal that the tolerance error for the density and pressure are $\pm 7.8\%$ and $\pm 3.4\%$, respectively.

Fig. 15 presents the impact of model input quantities of interest $X_i = \{\text{inlet flow rate, density, dynamic viscosity, outlet pressure}\}$, on output quantities of interest $Y = \{\text{pressure}\}$. All input parameters are perturbed

by 1%, 2% and 5% from their nominal values one by one, and their impacts are quantified on output OQI. For the sake of simplicity, here we only presented the impact of 2% change of each input quantities on output one, using the following sensitivity metrics:

$$S_i = \left| \frac{\partial Y}{\partial X_i} \right| \cong \left| \frac{Y(X_i + \Delta X_i) - Y(X_i)}{\Delta X_i} \right|. \quad (48)$$

where S_i is the magnitude of sensitivity index for model output with respect to i th input parameters. It is noticed that the dynamic viscosity is the most influential parameter for the inlet pressure along the pipeline, while, the outlet pressure is the least critical parameter.

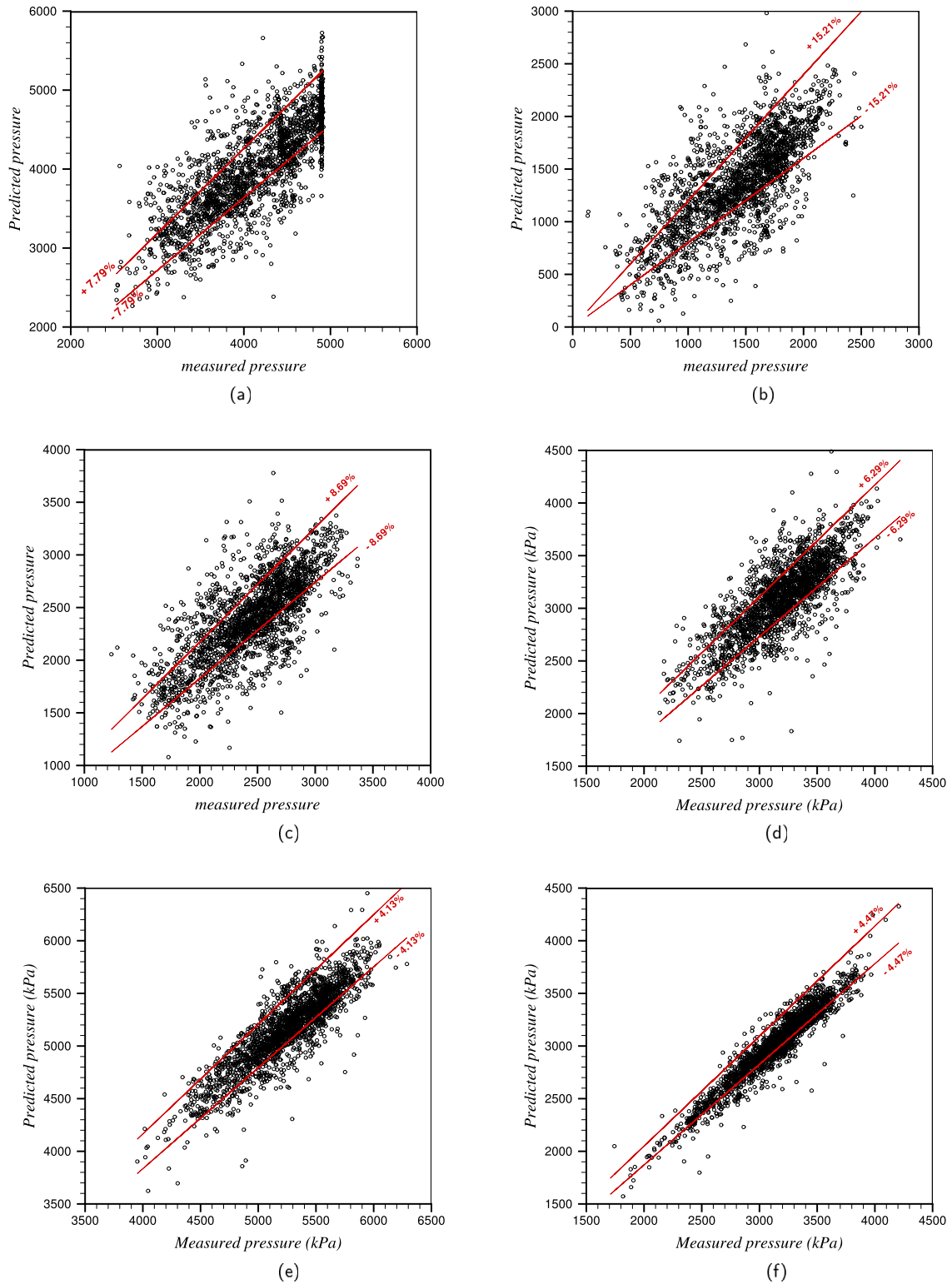


Fig. 11. Deviation of pressure monitoring sensor data (x-axis) and computed pressure (y-axis) at (a) Input main pipeline, (b) PMS-1, (c) Valve station, (d) PMS-2, (e) PMS-3 and (f) PMS-4.

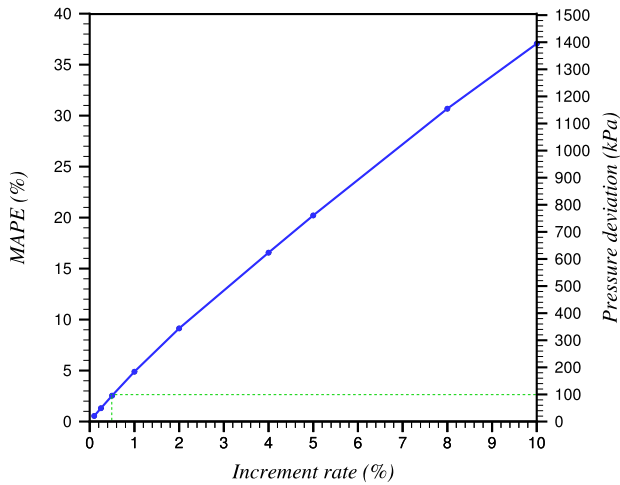


Fig. 12. Effect of flow rate uncertainty on the MAPE and the pressure deviation.

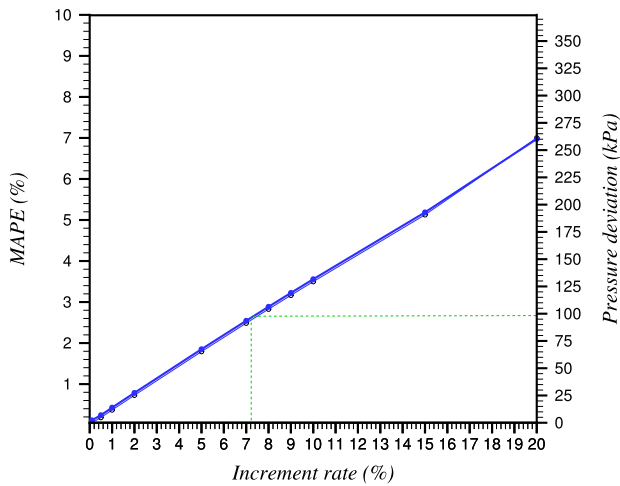


Fig. 13. Effect of increments in dynamic viscosity on the MAPE and pressure deviation.

7. Conclusion

In the present work, the dynamic phosphate slurry flow characteristics, i.e., flow rate, density, dynamic viscosity and pressure, through large-scale slurry pipelines, specifically the OCP Group's main pipeline in Morocco of length 187.124 km, are investigated by considering the real operating conditions. A dynamic model uses a section-averaged approach and the Finite Volume Method (FVM) is carried out to simulate, around 130 h of continuous flow (including downtime), the incompressible and non-Newtonian fluid phosphate slurry flow and predict its characteristics. The model's accuracy is validated against on-site measured data. The main conclusions from this work can be summarized as follows:

- The computed density at the outlet main pipeline is almost identical to the measured values, with an average deviation of $\pm 0.32\%$. However, a significant difference is observed in the downtime period.
- The present model predicts with high accuracy and reasonable computational efficiency the pressure evolution along the pipeline. The numerical measurements were validated against the data from the six pressure monitoring sensors installed on-site, showing less than 10% deviation at six positions along the main pipeline which is acceptable accuracy.

- The outcomes of the present investigation reveal that the dynamic viscosity value considered (a measured value) has a great effect on the computed pressure values, according to the sensitivity analysis. A 7.29% relative error in viscosity measurement can lead to a 100 kPa deviation in inlet pressure predictions.

Applications The implemented digital twin can be used for real time monitoring of the pipe to predict dysfunctions, to understand the flow condition inside the pipe, to test new operating points (e.g. increasing or decreasing the flow rates or the slurry density), as well as for the dimensioning of future slurry pipeline installations.

CRedit authorship contribution statement

Ahmed Ja: Conducted the data analysis, Interpretation, Writing – original draft. **Saad Benjelloun:** Conceptualization, Conducted the data analysis, Interpretation, Writing – original draft. **Jean Michel Ghidaglia:** Conceptualization, Design of the study. **Faical Ait Lahbib:** Provided critical revisions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Annexe

Settling velocity (V_{sv}) of the phosphate slurry represents the Limit Deposit Velocity, below which the sedimentation of the particles starts to occur. The computation of the settling velocity begins with dividing the particle size distribution into bands denoted as i as shown in Table 3, these data are measured at OCP Group lab.

The first parameters to be calculated are the representative particle size and velocity of each band as follows:

$$d_i^* = \left(\frac{\rho_w (\rho_p - \rho_w) * g}{\mu_w^2} \right)^{\frac{1}{3}} d_i. \quad (49)$$

$$\begin{cases} \text{For fine particles, } d_i^* \leq 4 : V_{sv-i}^* = 0.0556 d_i^{*2}. \\ \text{For medium particles, } 4 < d_i^* < 70 : V_{sv-i}^* = 0.21 d_i^*. \\ \text{For coarse particles, } d_i^* \geq 70 : V_{sv-i}^* = 0.0556 d_i^{*0.5}. \end{cases} \quad (50)$$

d_i^* and d_i are the representative particle size and reel diameter of each particle in the band i . The settling velocity value for each particle band can be calculated as follows:

$$V_{sv-i} = V_{sv-i}^* \left(\frac{\mu_w g (\rho_p - \rho_w)}{\rho_w^2} \right)^{\frac{1}{3}}. \quad (51)$$

The obtained results are shown in Table 3. From Table 3, it is clear that the settling velocity of the particles is very small with respect to the average velocity of the flow (≈ 2 m/s) and can be neglected, as long as both heterogeneous and homogeneous transport are taken into account.

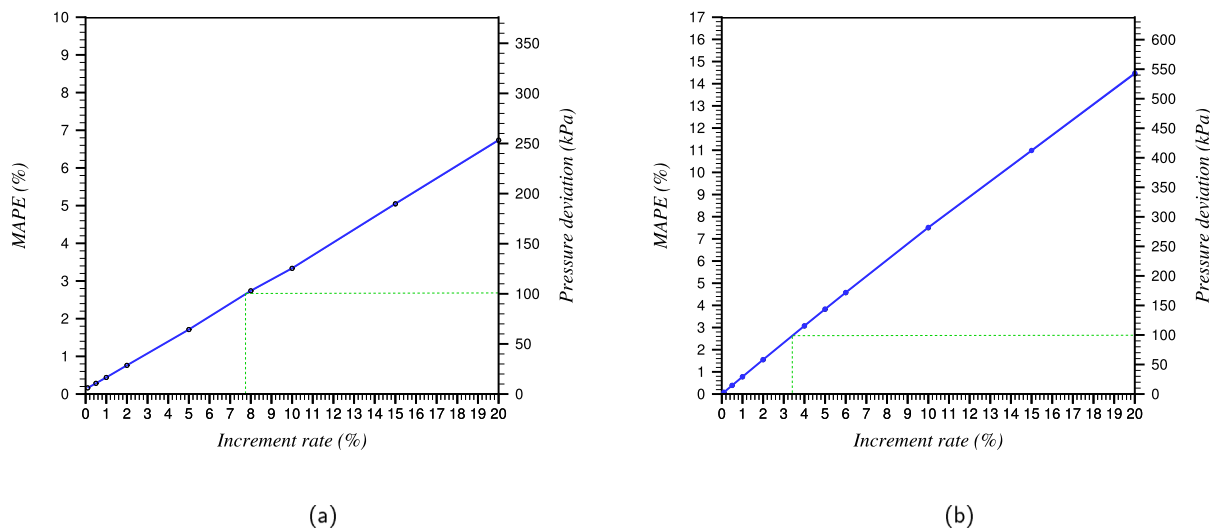


Fig. 14. Effect of increments in (a) inlet density and (b) outlet pressure on the MAPE and pressure deviation.

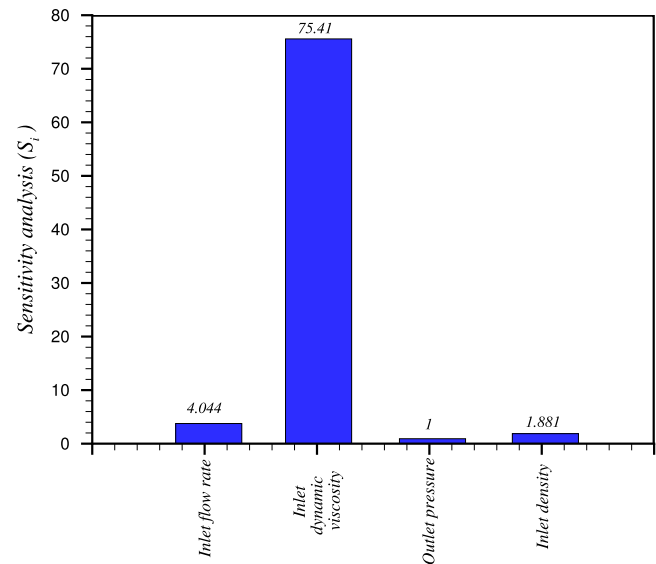


Fig. 15. Sensitivity analysis results of inlet pressure of the main pipeline.

Table 3
Particle size distribution and settling velocity of the Phosphate slurry.

Bands i	Particle size diameter	Proportion of particles	Settling velocity
1	<500 μm	99.88	6.86×10^{-2}
2	<400 μm	98.72	5.48×10^{-2}
3	<315 μm	95.06	4.32×10^{-2}
4	<250 μm	88.35	3.43×10^{-2}
5	<210 μm	81.29	2.88×10^{-2}
6	<160 μm	67.43	2.19×10^{-2}
7	<150 μm	63.89	2.09×10^{-2}
8	<125 μm	54.03	1.45×10^{-2}
9	<80 μm	34.79	5.94×10^{-3}
10	<74 μm	32.43	5.08×10^{-3}
11	<53 μm	25.72	2.61×10^{-3}
12	<44 μm	23.64	1.80×10^{-3}
13	<40 μm	22.82	1.48×10^{-3}
14	<37 μm	22.26	1.27×10^{-3}
15	<10 μm	15.71	9.27×10^{-5}

References

- Belbsir, H., El Hami, K., Soufi, A., 2018. Study of the rheological behavior of the phosphate water slurry and search for a suitable model to describe its rheological behavior. *Int. J. Mech. Mech. Eng.* 18, 73–81.
- Benjelloun, S., Desvillettes, L., Ghidaglia, J.-M., Nielsen, K., 2012. Modeling and simulation of thick sprays through coupling of a finite volume Euler equation solver and a particle method for a disperse phase. *Note Mat.* 32, 63–85.
- Bordet, A., Poncet, S., Poirier, M., Galanis, N., 2018. Flow visualizations and pressure drop measurements of isothermal ice slurry pipe flows. *Exp. Therm Fluid Sci.* 99, 595–604.
- Carravetta, A., Conte, M.C., Fecarotta, O., Martino, R., 2016. Performance of slurry flow models in pressure pipe tests. *J. Hydraul. Eng.* 142 (2).
- Dong, H., Aziz, N., Shafri, H., Ahmad, K., 2022. Numerical study on transportation of cemented paste backfill slurry in bend pipe. *Processes* 10 (8).
- Doron, P., Granica, D., Barnea, D., 1987. Slurry flow in horizontal pipes experimental and modeling. *Int. J. Multiph. Flow* 13, 535–547.
- Durand, R.-J., 1953. Basic relationships of the transportation of solids in pipes-experimental research. In: *Proc. IAHR 5th Congress, Minneapolis*. pp. 89–103.
- El Hamra, F., Boukharfane, R., Benjelloun, S., Ja, A., Ghidaglia, J.-M., 2022. Modeling of phosphates slurry pipelines through dynamic non-Newtonian fluid model with modelica. In: *Proc. 32nd European Symposium on Computer Aided Process Engineering*. pp. 415–420.
- El Moçayd, N., Seaid, M., 2021. Data-driven polynomial chaos expansions for characterization of complex fluid rheology: Case study of phosphate slurry. *Reliab. Eng. Syst. Saf.* 216 (107923).
- Elkarii, M., 2023. Towards modeling phosphate slurry flow using CFD technique (Ph.D. thesis). Université Paris Sciences et Lettres, in collaboration with Mohammed VI Polytechnic University, Paris, France and Benguerir, Morocco, Ph.D. Thesis.
- Elkarii, M., Boukharfane, R., Benjelloun, S., Bouallou, C., 2023a. A CFD-based surrogate model for predicting slurry pipe flow pressure drops. *Particul. Sci. Technol.* 41 (3), 432–442.
- Elkarii, M., Boukharfane, R., Benjelloun, S., Bouallou, C., El Moçayd, N., 2023b. Global sensitivity analysis for phosphate slurry flow in pipelines using generalized polynomial chaos. *Phys. Fluids* 35 (6), 063323.
- Geissler, B., Hermann, L., C. Mew, M., Steiner, G., 2018. Striving toward a circular economy for phosphorus: The role of phosphate rock mining. *Minerals* 8.
- Gidaspow, D., 1994. *Multiphase Flow and Fluidization*. Academic Press.
- Gómez González, R., Garzó, V., 2022. Kinetic theory of granular particles immersed in a molecular gas. *J. Fluid Mech.* 943.
- Grozdek, M., Khodabandeh, R., Lundqvist, P., 2009. Experimental investigation of ice slurry flow pressure drop in horizontal tubes. *Exp. Therm Fluid Sci.* 33 (2), 357–370.
- Han, J., Moraga, C., 1995. The influence of the sigmoid function parameters on the speed of backpropagation learning. In: *Mira, J., Sandoval, F. (Eds.), From Natural To Artificial Neural Computation*. In: *Lecture Notes in Computer Science*, vol. 930, Springer, Berlin, Heidelberg, pp. 195–201. http://dx.doi.org/10.1007/3-540-59497-3_175.
- Idelchik, I.E., 1986. *Handbook of Hydraulic Resistance*. Washington.
- Jacobs, B.E., 1991. *Design of Slurry Transport Systems*. Crc Press.
- Ling, J., Skudarnov, P., Lin, C., Ebadian, M., 2003. Numerical investigations of liquid-solid slurry flows in a fully developed turbulent flow region. *Int. J. Heat Fluid Flow* 24 (3), 389–398.

- Liu, S., Song, M., Hao, L., Wang, P., 2021. Numerical simulation on flow of ice slurry in horizontal straight tube. *Front. Energy* 15 (1), 201–207.
- Louge, M., Mastorakos, E., Jenkins, J., 1991. The role of particle collisions in pneumatic transport. *J. Fluid Mech.* 231, 345–359.
- Lun, C., Savage, S., Jeffrey, D., Chepur, N., 1984. Kinetic theory for granular flow: inelastic particles in Couette flow and slightly inelastic particles in a general flow field. *J. Fluid Mech.* 140, 223–256.
- Matoušek, V., 2002. Pressure drops and flow patterns in sand-mixture pipes. *Exp. Therm. Fluid Sci.* 26, 693–702.
- Messa, G.V., Yang, Q., Adediji, O.E., Chára, Z., Duarte, C.A.R., Matoušek, V., Rasteiro, M.G., Sanders, R.S., Silva, R.C., de Souza, F.J., 2021. Computational fluid dynamics modelling of liquid-solid slurry flows in pipelines: State-of-the-art and future perspectives. *Processes* 9 (9).
- Niezgoda-Zelasko, B., Zalewski, W., 2006. Momentum transfer of ice slurry flows in tubes, modeling. *Int. J. Refrig.* 29 (3), 429–436.
- Plass, L., Molerus, O., 1974. Simultane Wärmeübergangs- und Druckverlustmessungen an Feststoff/Flüssigkeits-Suspensionen im Übergangsbereich vom homogenen zum heterogenen Suspensionstransport. *Chem. Ing. Tech.* 46, 355.
- Roco, M.C., Cader, T., 1990. Energy approach for wear distribution in slurry pipelines. *Jpn. J. Multiphase Flow* 4, 2–20.
- Rusconi, J., Lakhouaja, A., Kopuz, M., 2016. The design and engineering of the 187 km khouribga to jorf lasfar phosphate slurry pipeline. *Procedia Eng.* 138, 142–150.
- Shi, B., Chen, Y., Wang, X., Song, S., Fu, S., Zhou, J., Liu, Y., Lv, X., Gong, J., Liu, Y., 2022. Flowloop investigation into hydrate formation and slurry flow in the presence of micron-sized sand particles. *J. Pet. Sci. Eng.* 212, 110251.
- Singh, J.P., Kumar, S., Mohapatra, S.K., 2021. Simulation and optimization of coal-water slurry suspension flow through 90° pipe bend using CFD. *Int. J. Coal Prep. Util.* 41 (6), 428–450.
- Sun, B., Fu, W., Wang, Z., Xu, J., Chen, L., Wang, J., Zhang, J., 2022. Characterizing the rheology of methane hydrate slurry in a horizontal water-continuous system. *SPE J.* 25 (3), 1026–1041.
- Vlasák, P., Chára, Z., Matoušek, V., Konfršt, J., Kesely, M., 2019. Experimental investigation of fine-grained settling slurry flow behaviour in inclined pipe sections. *J. Hydrol. Hydromech.* 67 (2), 113–120.
- Wang, J., Wang, S., Zhang, T., Liang, Y., 2013. Numerical investigation of ice slurry isothermal flow in various pipes. *Int. J. Refrig.* 36 (1), 70–80.
- Wasp, E.J., Aude, T.C., Kenny, J.P., Seiter, R.H., Jacques, R.B., 1970. Deposition velocities, transition velocities and spatial distribution of solids in slurry pipelines. In: *Proc. of the 1st International Conference on the Hydraulic Transport of Solids in Pipes*. pp. 53–76.
- Zhou, M., Kuang, S., Xiao, F., Luo, K., Yu, A., 2021. CFD-dem analysis of hydraulic conveying bends: Interaction between pipe orientation and flow regime. *Powder Technol.* 392, 619–631.
- Zisselmar, R., Molerus, O., 1979. Investigation of solid-liquid pipe flow with regard to turbulence modification. *Chem. Eng. J.* 18, 233–239.